

ForensicNova

Volatile memory forensic report

Acquisition identity

Acquisition ID	6a8728ed-c689-4e69-ac3d-51319d926afe
Operator	dfir-tester
Target VM	win2022-dfir-test-01
VM UUID	b995bb7a-f35f-4aed-a275-3701921c0ee1
Libvirt domain	instance-00000001
Acquisition started	2026-06-02T15:48:02.327140Z
Acquisition completed	2026-06-02T15:49:18.627444Z
Duration	76.3 s
MD5	662cfb7595128756db3a491507da77ac
SHA-1	5fb6050728c27cef86e635b9d8b2e1337edf9f22
Integrity verified	yes
Tool	ForensicNova v0.1.0
Report schema	1.2

Document generation

PDF generated at (UTC)	2026-07-08T18:49:13Z
Generated by	dfir-tester
Document type	Single-acquisition forensic report

Operator signature

Full name (block capitals)

authenticated as Keystone user: dfir-tester

Date and place

Signature

Evidence

Object name	dump-win2022-dfir-test-01-20260602T154802Z.raw
Swift container	forensics
Size	4.01 GiB (4,303,554,739 bytes)
Format	raw
Acquisition method	libvirt-coreDumpWithFormat
MD5	662cfb7595128756db3a491507da77ac
SHA-1	5fb6050728c27cef86e635b9d8b2e1337edf9f22
Swift ETag	6110dc6ba1683f8ab5e26cdd1b09f4fb
ETag verified	yes
Upload method	slo

SLO segments

#	Name	Size	MD5 / ETag
1	dump-win2022-dfir-test-01-20260602T154802Z.raw/seg-0001	4.00 GiB (4,294,967,296 bytes)	074bdc347ff0cd496a89920889213524
2	dump-win2022-dfir-test-01-20260602T154802Z.raw/seg-0002	8.19 MiB (8,587,443 bytes)	fcbee132498f38d873dd800835b6a4b5

The MD5 / ETag column shows the MD5 of each segment; for a segment uploaded in a single PUT the Swift ETag equals that MD5, so the two coincide. A differing ETag would be listed explicitly.

Target system

Hypervisor-side metadata. Useful for Volatility 3 profile / ISF selection. No data is ever read from inside the guest.

Nova

ID	b995bb7a-f35f-4aed-a275-3701921c0ee1
Name	win2022-dfir-test-01
Status	ACTIVE
Created	2026-05-26T10:23:27Z
Compute host	devstack
Hypervisor host	devstack

Flavor

ID	3
Name	m1.medium
vCPUs	2
RAM	4096 MB
Disk	40 GB

Glance image

ID	961d36a8-a832-4a0a-8d96-d3ef4ccd930f
Name	win2022-dfir-test
Disk format	qcow2
Container format	bare
OS type	windows
OS distro	windows
OS version	—
Architecture	—
HW machine type	—

Hypervisor & libvirt

Hypervisor type	kvm
Domain name	instance-00000001
Architecture	x86_64
Machine type	pc-i440fx-noble
CPU mode	custom
Memory	4096 MB
vCPUs	2

Chain of custody

14 events recorded by the acquisition pipeline. Persisted append-only on the hypervisor at /var/log/forensicnova/chain-of-custody.jsonl.

#	Timestamp	Event	Description
01	15:48:02.327	async_job_started	(no description registered for this event type)
02	15:48:02.329	acquisition_initiated	Acquisition pipeline started — instance identified
03	15:48:02.332	domain_lookup_completed	Nova UUID resolved to libvirt domain
04	15:48:02.334	memory_dump_started	Memory acquisition started — libvirt coreDumpWithFormat invoked
05	15:48:09.554	memory_dump_completed	Raw memory dump written to hypervisor staging area
06	15:48:09.570	dump_ownership_fixed	Dump file chowned to service user for pipeline access
07	15:48:09.575	hashing_started	MD5 + SHA-1 streaming hash started
08	15:48:27.913	hashing_completed	Hashes computed (single-pass, 64 KB chunks)
09	15:48:28.178	slo_upload_started	SLO upload started — file split into segments
10	15:49:01.491	swift_segment_uploaded	Segment uploaded and per-segment ETag verified locally
11	15:49:01.569	swift_segment_uploaded	Segment uploaded and per-segment ETag verified locally
12	15:49:01.609	swift_manifest_uploaded	SLO manifest PUT — Swift validated all segments server-side
13	15:49:01.609	swift_slo_upload_verified	SLO composite ETag verified end-to-end
14	15:49:18.626	local_dump_secure_deleted	Local dump shred-overwritten and unlinked

Analysis

5 analysis record(s) associated with this acquisition: 2 Volatility, 3 MISP enrichment.

Volatility analyses

Volatility · preset «fast» · 2026-06-02 15:53 UTC

Preset	fast
Volatility	2.28.0
OS hint	windows
Duration	203.722 s
Coherence	hashes match report
Analyzer MD5	662cfb7595128756db3a491507da77ac
Analyzer SHA-1	5fb6050728c27cef86e635b9d8b2e1337edf9f22

Plugins executed: 11 ok, 0 failed, 0 timeout, 0 parse error

Plugin	Status	Rows	Duration	Error
windows.cmdline.CmdLine	ok	104	1.89	—
windows.cmdscan.CmdScan	ok	3	23.85	—
windows.consoles.Consoles	ok	3	29.95	—
windows.getsids.GetSIDs	ok	1278	5.84	—
windows.info.Info	ok	21	1.22	—
windows.malware.malfind.Malfind	ok	5	76.37	—
windows.malware.psxview.PsXView	ok	115	15.66	—
windows.netscan.NetScan	ok	60	8.73	—
windows.pslist.PsList	ok	104	1.42	—
windows.pstree.PsTree	ok	5	1.58	—
windows.svcscan.SvcScan	ok	1158	22.46	—

Volatility · preset «full» · 2026-06-02 16:09 UTC

Preset	full
Volatility	2.28.0
OS hint	windows
Duration	783.11 s
Coherence	hashes match report
Analyzer MD5	662cfb7595128756db3a491507da77ac
Analyzer SHA-1	5fb6050728c27cef86e635b9d8b2e1337edf9f22

Plugins executed: 34 ok, 0 failed, 0 timeout, 0 parse error

Plugin	Status	Rows	Duration	Error
windows.amcache.Amcache	ok	395	30.65	—

windows.callbacks.Callbacks	ok	152	7.24	—
windows.cmdline.CmdLine	ok	104	1.59	—
windows.cmdscan.CmdScan	ok	3	21.34	—
windows.consoles.Consoles	ok	3	30.82	—
windows.dlllist.DllList	ok	4206	10.19	—
windows.driverirp.DriverIrp	ok	3342	8.53	—
windows.getsids.GetSIDs	ok	1278	4.94	—
windows.handles.Handles	ok	30942	57.20	—
windows.info.Info	ok	21	1.20	—
windows.malware.direct_system_calls.DirectSystemCalls	ok	24	63.73	—
windows.malware.hollowprocesses.HollowProcesses	ok	0	82.27	—
windows.malware.indirect_system_calls.IndirectSystemCalls	ok	0	118.88	—
windows.malware.ldrmodules.LdrModules	ok	4439	21.98	—
windows.malware.malfind.Malfind	ok	5	67.12	—
windows.malware.pebmasquerade.PebMasquerade	ok	104	1.86	—
windows.malware.processghosting.ProcessGhosting	ok	0	25.41	—
windows.malware.psxview.PsXView	ok	115	15.72	—
windows.malware.suspicious_threads.SuspiciousThreads	ok	2	84.18	—
windows.malware.svcdiff.SvcDiff	ok	0	22.71	—
windows.malware.unhooked_system_calls.UnhookedSystemCalls	ok	45	20.55	—
windows.modscan.ModScan	ok	145	5.36	—
windows.modules.Modules	ok	137	1.35	—
windows.netscan.NetScan	ok	60	7.69	—
windows.netstat.NetStat	ok	51	1.89	—
windows.pslist.PsList	ok	104	1.57	—
windows.psscan.PsScan	ok	117	6.63	—
windows.pstree.PsTree	ok	5	1.82	—
windows.registry.hashdump.Hashdump	ok	5	3.77	—
windows.registry.lsadump.Lsadump	ok	3	4.02	—
windows.registry.scheduled_tasks.ScheduledTasks	ok	220	12.36	—
windows.registry.userassist.UserAssist	ok	9	3.19	—
windows.ssdtd.SSDT	ok	948	1.88	—
windows.svcscan.SvcScan	ok	1158	19.12	—

MISP enrichment

MISP enrichment · 2026-06-02 15:55 UTC

THREAT SCORE: YELLOW

2 MISP match(es), no high-risk attribution

Injection signals

Scoring detectors — these determine the threat score
(precise detectors not run — full preset only)

Informational — context only, not scored

Malfind regions	5
Psxview-only (psscan)	6

Intelligence source	CIRCL OSINT Feed (CIRCL)
MISP events at query	1, 610
Coherence	hashes match report
IOCs extracted	107
IOCs checked	8
IOCs with match	2
IOCs filtered	99
ATT&CK techniques	Code Signing (T1553.002)

Indicators of Compromise (8)

Type	Value	Match	Attribution	Context
filename	mimikatz.exe	1	ATT&CK: T1553.002 Source: CIRCL	PID 3536, PPID 1660
filename	certutil.exe	1	Source: CIRCL	PID 3340, PPID 1660
ip-dst	203.0.113.42	0	—	ForeignAddr:4444 from PID 2288 (rundll32.exe) TCPv4 CLOSED
ip-dst	198.51.100.50	0	—	ForeignAddr:80 from PID 3340 (certutil.exe) TCPv4 SYN_SENT
filename	CompatTelRunne	0	—	PID 3008, PPID 1612
filename	payload.exe	0	—	PID 4020, PPID 1660
filename	rundll32.exe	0	—	PID 2288, PPID 4020
url	http://198.51.100.50/ stage2.bin	0	—	cmdline of certutil.exe (PID 3340)

MISP enrichment · 2026-06-02 16:09 UTC

THREAT SCORE: RED

precise injection detector(s) fired: suspicious_threads=2

Injection signals

Scoring detectors — these determine the threat score

Suspicious threads — fired	2
Hollow processes	0
Process ghosting	0
Service diff	0
Informational — context only, not scored	
Malfind regions	5
Psxview-only (psscan)	6
Intelligence source	CIRCL OSINT Feed (CIRCL)
MISP events at query	1,610
Coherence	hashes match report
IOCs extracted	107
IOCs checked	8
IOCs with match	2
IOCs filtered	99
ATT&CK techniques	Code Signing (T1553.002)

Indicators of Compromise (8)

Type	Value	Match	Attribution	Context
filename	mimikatz.exe	1	ATT&CK: T1553.002 Source: CIRCL	PID 3536, PPID 1660
filename	certutil.exe	1	Source: CIRCL	PID 3340, PPID 1660
ip-dst	203.0.113.42	0	—	ForeignAddr:4444 from PID 2288 (rundll32.exe) TCPv4 CLOSED
ip-dst	198.51.100.50	0	—	ForeignAddr:80 from PID 3340 (certutil.exe) TCPv4 SYN_SENT
filename	CompatTelRunne	0	—	PID 3008, PPID 1612
filename	payload.exe	0	—	PID 4020, PPID 1660
filename	rundll32.exe	0	—	PID 2288, PPID 4020
url	http://198.51.100.50/stage2.bin	0	—	cmdline of certutil.exe (PID 3340)

MISP enrichment · 2026-07-08 18:42 UTC

THREAT SCORE: RED

high-risk galaxy match: Tool:Meterpreter

Injection signals

Scoring detectors — these determine the threat score

Suspicious threads — fired	2
Hollow processes	0
Process ghosting	0
Service diff	0

Informational — context only, not scored

Malfind regions	5
Psxview-only (psscan)	6
Intelligence source	CIRCL OSINT Feed (CIRCL)
MISP events at query	1,647
Coherence	hashes match report
IOCs extracted	107
IOCs checked	8
IOCs with match	5
IOCs filtered	99
Threat actors	UNC-FNDEMO (simulated)
Tools	Meterpreter, Mimikatz
ATT&CK techniques	OS Credential Dumping (T1003), Process Injection (T1055), Application Layer Protocol (T1071), Ingress Tool Transfer (T1105), Code Signing (T1553.002)

Indicators of Compromise (8)

Type	Value	Match	Attribution	Context
filename	mimikatz.exe	2	Tool: Mimikatz ATT&CK: T1003 · T1553.002 Source: CIRCL, ForensicNova	PID 3536, PPID 1660
ip-dst	203.0.113.42	1	Actor: UNC-FNDEMO (simulated) Tool: Meterpreter ATT&CK: T1055 · T1071 · T1105 Source: ForensicNova	ForeignAddr:4444 from PID 2288 (rundll32.exe) TCPv4 CLOSED
ip-dst	198.51.100.50	1	Actor: UNC-FNDEMO (simulated) Tool: Meterpreter ATT&CK: T1055 · T1071 · T1105 Source: ForensicNova	ForeignAddr:80 from PID 3340 (certutil.exe) TCPv4 SYN_SENT
filename	certutil.exe	1	Source: CIRCL	PID 3340, PPID 1660
url	http://198.51.100.50/stage2.bin	1	Actor: UNC-FNDEMO (simulated) Tool: Meterpreter ATT&CK: T1055 · T1071 · T1105 Source: ForensicNova	cmdline of certutil.exe (PID 3340)
filename	CompatTelRunne	0	—	PID 3008, PPID 1612
filename	payload.exe	0	—	PID 4020, PPID 1660
filename	rundll32.exe	0	—	PID 2288, PPID 4020

Notice

This report is generated automatically by the ForensicNova plugin from the canonical JSON records of a single acquisition: the acquisition report and the associated Volatility 3 and MISP analysis records produced by the ForensicNova analyzer. Those JSON records, the raw memory image, and the chain-of-custody journal are the primary evidentiary artifacts; this PDF is a printable presentation of them intended for legal hand-off and operator countersignature.

Cryptographic integrity of the memory image is established at the hypervisor before any transfer (MD5 + SHA-1, streaming) and verified end-to-end against Swift via ETag comparison. Before processing, each analysis independently re-reads the stored image and confirms its hashes against the acquisition record (the coherence check reported in every analysis section). The chain of custody is appended to a journal on the hypervisor and embedded in the JSON report. Any tampering with stored evidence breaks the hash match; tampering with the PDF does not affect the underlying evidence.

Forensic analysis is performed by the ForensicNova analyzer service: memory extraction with Volatility 3 and automated indicator enrichment against MISP threat intelligence. Threat-intelligence correlations are only as authoritative as the sources behind them; the feeds queried at analysis time are recorded in each enrichment (the Intelligence source field), and every reported match should be read against that provenance.

Each generation of this PDF is a distinct act and carries the UTC timestamp of generation. The operator named on the cover is the analyst who performed the acquisition and the subsequent analysis, and who countersigns this document. Reproducing this PDF at a later time will produce a new, byte-different document with a new generation timestamp; the underlying evidence and analysis records remain the same.

ForensicNova is an academic prototype developed at the Università degli Studi di Salerno (ISISLab) as the M.Sc. thesis project for the degree programme in Sicurezza Informatica e Tecnologie Cloud, having originated as coursework for the Piattaforme di Cloud Computing course. Use in production forensic workflows is the responsibility of the deploying organisation.